

NYDA, Russian Federation

WMO: 233450

Lat: 66.624N

Lon: 72.896E

Elev: 24

StdP: 14.68

Time Zone: 5.00 (E05)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-45.0	-40.1	-50.0	0.3	-44.2	-45.6	0.4	-39.7	34.8	7.7	29.7	10.2	6.2	130	0.942

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.8	80.3	64.9	75.9	63.6	70.4	60.6	66.7	75.6	64.3	73.8	61.9	69.9	11.8	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.0	86.2	72.0	60.4	78.4	68.9	57.9	71.7	66.2	31.3	76.5	29.5	74.0	27.8	70.0	75.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 31.9	27.6	24.6	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(5)			WB	-47.6	69.2	3.3	3.3	-49.9	71.6	-51.9	73.6	-53.7	75.5	-56.1	77.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	21.9	-12.6	-7.0	4.2	16.3	27.8	49.9	60.0	52.7	42.0	24.9	4.3	-2.2	
	DBStd	26.98	16.29	16.57	16.72	11.27	9.28	10.91	8.12	5.93	6.64	9.39	13.78	16.47	
	HDD50	10859	1941	1596	1419	1010	690	141	8	37	252	778	1370	1617	
	HDD65	15800	2406	2016	1884	1460	1153	461	202	383	690	1243	1820	2082	
	CDD50	595	0	0	0	0	2	139	320	121	13	0	0	0	
	CDD65	60	0	0	0	0	0	9	49	2	0	0	0	0	
	CDH74	555	0	0	0	0	0	84	457	14	0	0	0	0	
	CDH80	111	0	0	0	0	0	10	101	0	0	0	0	0	
Wind	WSAvg	12.3	11.9	11.9	13.2	13.0	13.7	12.2	11.6	11.8	12.2	12.1	12.2	11.8	
Precipitation	PrecAvg	13.9	0.6	0.6	0.7	0.9	1.1	1.5	2.0	1.9	1.6	1.5	0.9	0.8	
	PrecMax	19.3	1.5	2.6	1.6	2.8	2.1	3.1	5.3	3.1	2.6	2.8	1.8	1.9	
	PrecMin	7.9	0.2	0.2	0.1	0.0	0.3	0.4	0.1	0.7	0.4	0.8	0.4	0.3	
	PrecStd	2.8	0.3	0.4	0.4	0.6	0.5	0.6	1.2	0.7	0.6	0.5	0.4	0.4	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.1	27.7	35.1	38.6	61.5	81.3	86.4	76.2	66.0	46.6	35.0	30.3	
		MCWB	26.6	26.5	33.3	34.6	52.9	65.0	66.8	62.2	55.9	45.2	34.1	29.5	
	2%	DB	19.7	23.6	32.4	36.1	50.5	76.2	82.7	70.3	58.5	41.4	31.5	26.5	
		MCWB	18.8	22.3	30.8	33.7	44.6	61.8	65.5	60.1	52.9	39.1	30.7	25.4	
	5%	DB	15.7	20.1	30.3	34.8	42.9	71.1	79.5	65.7	55.3	38.4	28.2	23.6	
		MCWB	15.0	19.0	28.8	32.9	39.2	58.9	65.0	58.3	50.7	36.4	27.4	22.8	
	10%	DB	10.6	15.7	27.8	33.1	37.8	67.2	75.1	62.5	51.8	35.6	23.8	20.4	
		MCWB	9.9	14.8	26.5	31.6	35.3	58.5	64.4	56.3	48.8	34.1	22.8	19.6	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.0	26.9	33.3	35.7	53.0	67.0	71.1	64.5	57.1	45.1	33.8	29.7
			MCDB	27.5	27.9	34.5	37.3	61.7	77.0	82.6	72.0	63.8	47.1	34.8	30.0
2%		WB	19.0	22.4	31.1	34.1	45.6	64.4	68.8	62.0	54.0	39.7	30.8	25.8	
		MCDB	19.7	23.5	32.3	35.7	50.6	72.5	78.5	68.6	58.0	40.8	31.5	26.6	
5%		WB	15.0	19.1	28.9	33.1	39.6	61.2	66.2	59.2	51.3	36.6	27.4	22.9	
		MCDB	15.7	20.1	30.2	34.6	42.7	69.1	75.4	65.0	54.3	37.9	28.1	23.6	
10%		WB	9.8	14.7	26.5	31.7	35.6	58.1	64.0	56.7	48.9	34.1	22.9	19.7	
		MCDB	10.4	15.6	27.9	33.3	37.4	66.4	74.1	61.2	51.6	35.3	23.7	20.4	
Mean Daily Temperature Range	5% DB	MDBR	13.3	14.4	15.4	16.2	10.8	13.8	13.8	11.2	9.4	8.8	12.8	14.0	
		MCDBR	18.3	15.7	13.7	13.3	18.1	23.1	23.2	18.8	16.1	9.9	9.6	13.2	
		MCWBR	18.1	15.5	13.1	11.7	12.6	12.0	10.3	10.2	10.6	7.6	9.6	13.1	
	5% WB	MCDBR	18.1	16.0	14.4	11.6	17.1	21.2	20.9	15.9	14.3	8.7	10.2	12.7	
		MCWBR	17.9	15.9	13.8	10.3	12.3	12.9	10.6	10.6	10.2	7.1	10.1	12.7	
Clear-Sky Solar Irradiance	taub		0.211	0.262	0.312	0.331	0.350	0.370	0.395	0.374	0.325	0.281	0.205	N/A	
	taud		1.871	2.011	2.015	2.056	2.151	2.344	2.324	2.362	2.447	2.214	1.923	N/A	
	Ebn at Noon		51	176	226	258	266	265	252	245	233	172	50	0	
	Edh at Noon		5	18	31	39	39	34	34	29	21	15	4	0	
All-Sky Solar Radiation	RadAvg		18	180	693	1372	1810	1691	1559	1053	578	209	42	0	
	RadStd		2	21	38	84	89	130	202	117	72	31	5	0	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page